

Customer Sales Data: Cleaning and EDA Report

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Tools: Python (Pandas, NumPy, Matplotlib, Seaborn), Jupyter Notebook

1. Project summary

This project takes a raw customer sales file with many data quality problems, cleans it, and explores it to answer basic business questions. The work was done in a Jupyter notebook (DataCleaning.ipynb) on one CSV file (messy_customer_sales_data.csv). In this report, "records" means rows in the file, because Customer_ID turned out not to identify one customer (see section 5).

Goals:

- Find and fix the quality problems in the raw data.
- Keep a clean dataset that can be trusted for analysis.
- Look at purchase amount by city, age group, gender and month.
- Calculate basic KPIs and a feedback split, and note what the data can and cannot tell us.

2. About the dataset

Item	Detail
Size	10,400 rows and 10 columns in the raw file. 10,197 rows after cleaning
Columns	Customer_ID, Name, Gender, Age, City, Signup_Date, Last_Purchase_Date, Purchase_Amount, Feedback_Score, Email
Signup dates	1 Jan 2018 to 31 Dec 2023
Last purchase dates	1 Jan 2023 to 31 Dec 2025
Cities	12 Indian cities (Chennai, Mumbai, Pune, Hyderabad, Bangalore, Jaipur, Delhi, Kolkata, Surat, Nagpur, Ahmedabad, Lucknow)

3. Problems found in the raw data

Problem	What was found
Duplicate rows	203 exact duplicate rows
Missing values	Name 504, Gender 495, Age 756, City 742, Last_Purchase_Date 591, Purchase_Amount 652, Feedback_Score 509, Email 655
Age stored as text	Values like "46 years" made the column text instead of numbers (1,453 values contained text)
Gender written many ways	male, female, m, f, MALE, FEM and so on, with mixed case and extra spaces
City spelling and case	Values like "ahmedabad" and "JAIPUR" with mixed case and spaces
Dates stored as text	Signup_Date and Last_Purchase_Date were read as plain text
Impossible dates	276 rows had a last purchase date earlier than the signup date

4. Cleaning steps

Step	Action taken
1	Pulled the number out of Age text using a regular expression, then set any age below 0 or above 100 to blank (none were found)
2	Converted both date columns to proper dates. Anything unreadable became blank (NaT)
3	Trimmed spaces and lowercased Gender, then mapped everything to Male or Female
4	Trimmed spaces and applied title case to City
5	Removed the 203 duplicate rows and reset the index (10,400 rows down to 10,197)
6	Filled missing Name, City, Gender and Email with "Unknown"
7	Checked emails (9,559 checked, none malformed), negative purchase amounts (none, minimum is 5,009.30) and feedback range (1 to 10)
8	Set the 276 impossible last purchase dates to blank and flagged them in a Date_Issue column
9	Created Age_Group with three equal-sized bands (Young 18 to 35, Adult 35 to 53, Senior 53 to 70). Missing ages were labelled "Unknown"
10	Left missing Age, Purchase_Amount, Feedback_Score and Last_Purchase_Date blank on purpose, so no made-up values enter the analysis
11	Checked outliers with box plots and the IQR rule. None were found in Age, Purchase_Amount or Feedback_Score

After cleaning, Gender has Female 4,856, Male 4,855 and Unknown 486. Remaining blanks: Age 743, Last_Purchase_Date 861, Purchase_Amount 640, Feedback_Score 504.

5. Data integrity issue: Customer_ID

The same Customer_ID is attached to different people. Ignoring rows where the name is "Unknown", 2,542 of 6,204 IDs (about 41%) have more than one name. One ID has as many as 6 different names, with different genders, ages and cities.

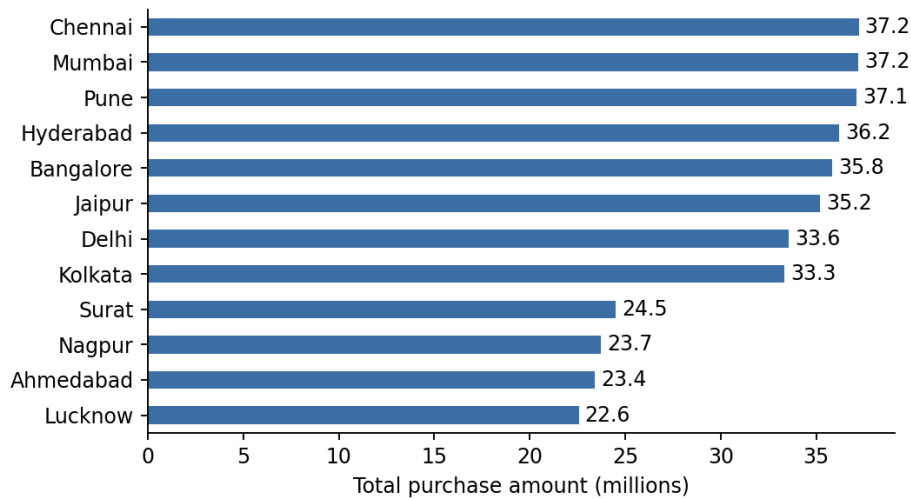
Example: CUST10004 appears as Mark Thomas in Pune (signed up 6 Aug 2020) and as Emily Adams in Ahmedabad (signed up 13 May 2021).

What this means:

- Customer_ID cannot be used to count unique customers.
- Repeat customer, retention and churn analysis are not possible with this file.
- For this reason the report does not give unique or repeat customer numbers.

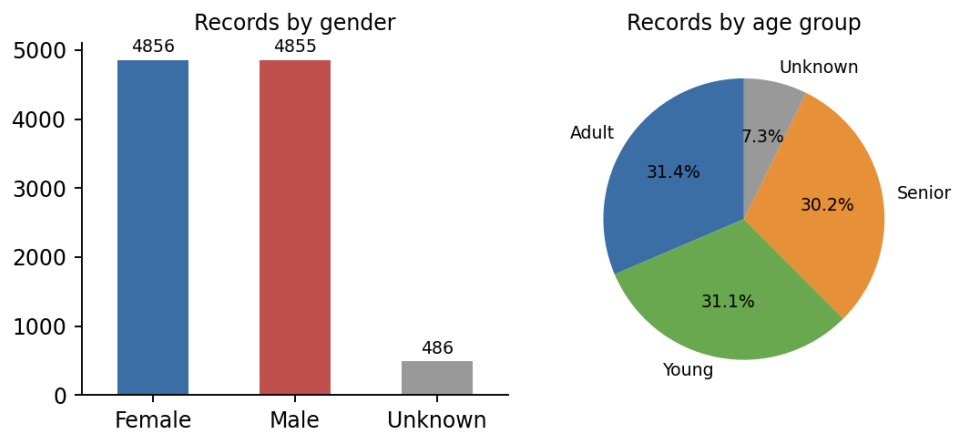
6. Analysis and charts

Purchase amount by city



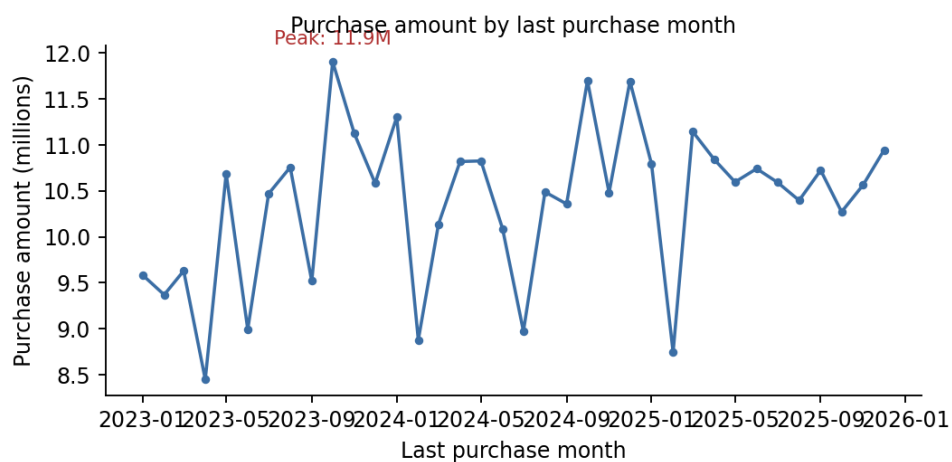
Rows with an unknown city were left out. The numbers are in millions.

Records by gender and age group



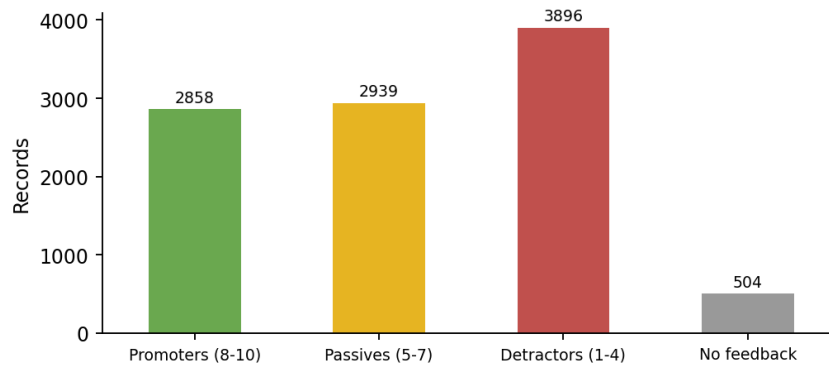
Male and female records are almost equal, and the three age groups are close in size.

Purchase amount by last purchase month



Each row has one amount and one last purchase date, so this shows when customers last bought, not real monthly sales.

Feedback groups



Custom bands: 8 to 10 Promoter, 5 to 7 Passive, 1 to 4 Detractor. These are not the standard NPS bands.

7. Business KPIs

KPI	Value
Total purchase amount	408,560,586.60
Average purchase amount	42,749.88
Highest and lowest purchase	79,997.70 and 5,009.30
Purchase amount by year (by last purchase date)	2023: 121.1M, 2024: 125.8M, 2025: 126.4M
Year on year growth	2024: +3.86%, 2025: +0.49%
Month on month growth	Swings between about -21% and +27%, with no steady direction
Average days from signup to last purchase	About 1,318 days (median 1,309)
Feedback (of 9,693 records with a score)	Detractors 40.2%, Passives 30.3%, Promoters 29.5%

8. Key findings

- **Data quality:** the raw file had 203 duplicate rows, 276 impossible dates, and messy age, gender and city values. All were fixed or flagged, leaving 10,197 usable rows.
- **Overall size:** total purchase amount is 408.6 million, with an average of about 42,750 per record. Amounts range from 5,009 to 79,998.
- **Two groups of cities:** eight cities have 33 to 37 million each (Chennai, Mumbai and Pune lead at about 37 million). Four cities are clearly lower: Surat, Nagpur, Ahmedabad and Lucknow, at 22.6 to 24.5 million each.
- **Why those four are lower:** they have fewer records (574 to 612, against 818 to 941 elsewhere). The average amount per record is almost the same in every city (about 41,600 to 43,900). So the gap comes from volume, not from lower spending.
- **Age and gender make no visible difference:** female and male records are equal in number, median purchase is about 42,800 to 43,600 across the age groups, and the correlation between age and amount is about -0.01.
- **Time trend is weak:** the monthly amount moves between 8.4 million (Apr 2023) and 11.9 million (Oct 2023) with no clear direction. Yearly amount grew 3.86% in 2024 and only 0.49% in 2025. Because this uses last purchase date, treat it as a rough signal only.

- **Feedback is mostly low:** 40.2% of scores fall in the Detractor band, more than Promoters (29.5%). Yet feedback score has no relation to purchase amount (correlation about -0.01), so unhappy records spend about as much as happy ones.
- **Customer_ID is not reliable:** about 41% of IDs belong to more than one name, so customer-level analysis cannot be trusted.